

Computer Vision for Ethnographic Research:
Finding Visual Predictors of Neighborhood Demographic Changes

A Thesis Presented to the Faculty of Architecture, Planning & Preservation

Columbia University

In Partial Fulfillment

Of the Requirements for the Degree

Master of Science in Urban Planning

By

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May 2023

Acknowledgments

Expressing my gratitude to those who have supported me throughout my academic journey feels like an understatement, as their contributions have been immeasurable. I am indebted to my advisor, Professor Anthony Vanky, for his invaluable guidance, unwavering support, and patience as I embarked on this project, and to my reader, Professor Sybil Wa for her encouragement and feedback. I'm thankful to David Mori for his critical contributions to the literature reviews used in this project. I would also like to express my gratitude to Columbia GSAPP and the University of Michigan Office of the Vice President for Research's Anti-Racism Grant program for their generous support of my research as part of the "Asians in Southeast Michigan: Information Networks and Placemaking as Responses to Racist Violence" project.

I am thankful beyond measure for my family, whose constant love, support, and sacrifices served as motivation when I hit plateaus. I am grateful to Professor Boyeong Hong for her inspiring classes and contagious enthusiasm for urban analytics which deepened my interest in the field. My dear friends, Lizzie Lee, Kit Nga Chou, Michelle Chen, and Smriti Muralishankar, provided much-needed emotional support and inspiration, and I am forever grateful for their unwavering friendship and encouragement. And, of course, I cannot forget my dear cat, Momo, who has been my constant companion and source of comfort through the long hours of research and writing. I am deeply grateful to everyone else who has played a role in my academic success, and I will forever cherish the memories of their support, encouragement, and kindness.

Abstract

This paper proposes a novel computer vision method for conducting large-scale and automated analysis of the commercial landscape of cities over time. The method employs ML techniques and leverages street view imagery to extract and analyze visual information, such as colors and languages used across storefronts, to identify changes in the characteristics of ethnic enclaves in cities. The paper argues that this method could be used in the context of ethnographic research to study the cultural practices and beliefs of communities in cities like Detroit, New York, and Los Angeles, and to develop a digital tool that maps the quantifiable changes in the characteristics of ethnic enclaves over time. By using qualitative data to draw causal inferences between these changes and events that may have had an impact, the tool could be used by urban planners and sociologists to better understand and predict urban change in the context of ethnic enclaves and cities. The paper also proposes a framework for incorporating this data into urban planning and sociological research, providing a basis for further study in this area.

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Introduction

In the past several decades, the United States has seen various waves of changes in the ethnic composition of the population, along with the formation and dissolution of unique enclaves. With the ever-growing population of our planet, modern cities have grown beyond the perceptual capacity of the human mind, posing problems for planners who need to gain a sociological understanding of cities to identify and solve key urban issues.

Preexisting datasets used by planners today have their uses but aren't without flaws. Census data tends to be aggregated, with ambiguous racial categories that vary over the decades. The updation frequency is also low. Land use data is another commonly referenced dataset, but like the Census data, it tends to be aggregated and with arbitrary categories. It also lacks cultural and ethnic contexts, which can only be known at a certain level of granularity. This level of granularity is achieved in Commerce data, which is discrete and more frequently updated. However, it overlooks informality and misses local contexts. There is a pressing need to overcome these gaps in data in order to allow planners to make better decisions through the employment of data tools.

Ethnography & Placemaking: Literature & Reflections

"As a result of living, working, and socializing in the spaces and places associated with "whiteness," the immigrants reinforce their invisible status within the larger sociopolitical community, though many newcomers preserve their ethnic identities through social and religious gatherings (*Contemporary Asian America (Third Edition)*, n.d.).

In a chapter of Min Zhou's book "Enclaves, Ethnoburbs, and New Patterns of Settlement Among Asian Immigrants," *Contemporary Asian America (Third Edition)*, Li argues that

globalization has shifted the landscape of immigration patterns in the US and Asia, bringing new waves of immigrants with different historical backgrounds to new parts of the US, namely, suburbia. The article digs somewhat deeper than Ethnoburb into the “ever-increasing heterogeneity within the immigrant community,” where individuals speaking different languages, coming from different classes, or practicing different religions can all be lumped in together under the label “Asian American.” Indeed, as Li’s piece points out, such diversity can even be found within a single ethnic group. Also of interest: Li attributes the lack of overt forms of political participation amongst these new immigrants to a “lack of understanding of American politics and/or their noncitizen status,” yet notes that they find other ways to get involved in local governance through participation in institutions such as school boards or business associations.

Min Zhou goes on to hypothesize that changes in the ethnic composition of a community may have important implications for the types of businesses that emerge and the opportunities available to entrepreneurs (Zhou, 2004). This thesis aims to test the converse of this hypothesis, i.e. whether through examining changes in the types of businesses, the languages spoken in advertising, the clientele served, etc, we can gain insights into the demographic shifts in a community.

This thesis proposes the usage of imagery to overcome the gaps identified in the other popular data sources as image data can provide valuable insights into urban contexts, allowing planners to gain a better understanding of the landscape, surroundings and quality of neighbourhoods. The higher frequency of updation, reduced ambiguity owing to its observational nature, and the granularity of the data makes it better equipped to be used as a tool to study change. Imagery can also be used in tandem with other data sources to

comprehensively document commerce in terms of typology, target demographic and industries that are prone to clustering. However, image data is notoriously difficult to digitally interpret— and at the city scale, the volume of discrete images can be overwhelmingly high for manual analysis.

Employing Artificial Intelligence to overcome these difficulties isn't a futuristic concept, but rather one that is already being employed. Inside the field of Artificial Intelligence, Computer Vision is a subfield that includes technologies allowing computers to "learn" to recognize a picture or the characteristics of images (Boesch, 2022). When used responsibly, computer vision can have a significant positive social impact by providing visibility to parameters that aren't often closely observed (How Computer Vision and Other Tech Is Enabling the Next Generation of Smart Cities - AI for Good Foundation, 2022), either due to a lack of resources or a lack of awareness.

Visual Indicators of Ethnographic Change in Commerce

Commerce can be seen as a mirror of the ethnographic changes in the areas surrounding its locale because they often reflect the needs and preferences of the local population. These changes can take many forms, such as changes in the types of businesses, the languages spoken in advertising, and the clientele served. By examining these changes, planners can gain insights into the demographic shifts taking place and make informed decisions about the development and revitalization of urban neighborhoods.

In his 2014 paper, *My Los Angeles: From Urban Restructuring to Regional Urbanization*, Edward W. Soja examines how the city of Los Angeles has undergone significant transformations in recent decades, and how these changes reflect broader shifts in the global

economy. Soja highlights the changing nature of the city's urbanization, which has resulted in the decentralization of urban functions and the emergence of new forms of governance, transportation networks, and suburban landscapes, all of which have had significant impacts on the city's ethnic landscape and commercial spaces.

The types of businesses that open in a neighborhood often reflect the economic status and purchasing power of the residents. For example, the emergence of high-end retail stores and luxury apartments often signals the arrival of affluent residents in a neighborhood. On the other hand, the presence of discount stores and pawn shops may indicate economic distress in a neighborhood. The emergence of new high-end retail stores in low-income neighborhoods can also lead to feelings of exclusion and resentment among long-time residents who are unable to afford the prices of the new businesses. Similarly, the closure of long-standing businesses that serve the needs of the original population can lead to a sense of loss and dislocation. In his paper, Soja argues in support of this statement that the postmetropolitan and regional urbanization era has resulted in the displacement of some ethnic businesses and the emergence of new forms of spatial segregation. He notes that "the postmetropolitan era is a period of urbanization beyond the metropolis, a shift from the old industrial city of factories and dense residential districts to a new, more fluid urban form based on decentralized networks of consumption and communication" – resulting in the emergence of new commercial spaces that cater to a different demographic than the traditional ethnic enclaves of the city, which in turn led to the displacement of some ethnic businesses. He highlights the growth of the high-tech industry in Los Angeles and the resulting development of new commercial spaces that serve a different demographic than the traditional ethnic enclaves of the city.

Wei Li's *Ethnoburb* is the foundational text in the study of the ethnoburb: a new form of the ethnic enclave that is formed outside of this history in suburbs such as Monterey Park, California, where there is a sizable Chinese American population. Classic theories of assimilation outline a process through which immigrant populations are "assimilated" into the mainstream white culture when they move into urban areas that bring them into close proximity with white norms and values. In contrast, the shifting tides of geopolitics and the ways they have affected push and pull factors bringing individuals to America, have provoked new waves of immigrants from countries in Asia that are not settling in the traditional urban enclaves, instead moving straight to the suburbs. Like traditional urban enclaves, these ethnoburbs allow immigrants the comfort of living around other immigrants and having access to familiar foods and institutions. Yet even with all of these comforts, there are still a number of problems encountered by residents who are discriminated against by longtime residents who feel their neighborhoods are being invaded. Li's work is especially important, as she highlights the heterogeneity to be found not just amongst Asian Americans, but in a given ethnic community made up of individuals who immigrated at different times and with different class and cultural backgrounds, differences which can, at times, lead to divisions.

Gentrification sees new residents bringing with them fresh requirements for goods & services thus generating newer bespoke shops targeting affluent white consumers whereby long-time establishments aimed at other demographics start giving way under increased pricing pressures favoring newcomers over locals leading eventually too many socio-cultural changes happening quickly enough when less profitable original enterprises shut down altogether or relocation occurs elsewhere thereby altering overall mood among communities causing tension through clashes arising between neighbors newly arrived versus those who

have been rooted longer than others feeling left behind or dislocated. This can in turn have a significant impact on the social and cultural dynamics of a neighborhood. The emergence of new businesses can bring new cultural practices and norms to a neighborhood, which can lead to tension and conflict between residents.

Colors and languages are essential factors in business promotion that are significantly influenced by cultural and ethnic contexts. It is important to understand the cultural nuances surrounding colors, as they can impact how they are perceived and used in marketing. For instance, in many Asian cultures, the color red is commonly associated with prosperity and fortune and is often used in advertising during festive periods (Elliot & Maier, 2014). On the other hand, in some African cultures, the color red can connote danger or warning and may not be the best choice for marketing (Johnson & Becker, 2010).

In the same vein, language choices can be influenced by cultural and ethnic contexts, and businesses must consider using the appropriate language to connect with specific communities effectively. Spanish, for instance, is an essential language for marketing to Hispanic communities in the United States since it is the country's second most spoken language. In Canada, French is also a critical language for businesses to reach out to Quebecois communities and comply with language laws (Bourhis & Giles, 1977). When a neighborhood experiences an influx of immigrants from a particular region, it is likely that storefronts will emerge to cater to the cultural and linguistic needs of that population. Ethnic enclaves are a prime example of this phenomenon, as they often contain businesses that cater to specific cultural groups. If the language used in signage and advertisements changes from Spanish to English, it may suggest that the demographic makeup of the community is shifting towards non-Hispanic populations. Conversely, if a previously monolingual area begins to

feature advertising in multiple languages, it may indicate an increase in immigrant populations.

The demographics of customers served by businesses can hint at ethnographic transitions in a locale. Min Zhou's 2004 paper, *Revisiting Ethnic Entrepreneurship: Convergences, Controversies, and Conceptual Advancements*, proposes a framework that identifies three main dimensions of ethnic entrepreneurship: cultural capital, social capital, and institutional context. Cultural capital refers to the resources, values, and norms that are specific to a particular ethnic group, whereas social capital includes the networks, relationships, and social resources that facilitate entrepreneurship, such as family support. Institutional context refers to the wider economic, political, and regulatory environment, such as immigration policies or discrimination laws.

Zhou reiterates that different ethnic groups may have different experiences and face different challenges. For example, African American entrepreneurs may face discrimination in obtaining funding or access to markets, while Asian American entrepreneurs may face cultural barriers to assimilation and integration. These communities also tend to have their own unique way of tackling these challenges. For instance, Asian communities increasing within once predominantly Hispanic areas might spawn new establishments that cater to them such as restaurants and grocery stores - the former perhaps specializing in unique dishes or fusions of different cuisines. Additionally, this surge may bring about hybrid enterprises which target multiple linguistic groups thereby reflecting society's evolving cultural landscape.

Additionally, iconography can play a critical role in communicating cultural and ethnic contexts. Using specific symbols or images can evoke particular emotions or ideas that can

resonate with a particular cultural or ethnic group. For example, employing traditional Chinese symbols in marketing to Chinese communities can help establish cultural relevance and forge a deeper connection with consumers. Similarly, using images of African American models in advertising can help businesses build trust and loyalty with Black consumers (Madden et al., 2000).

This thesis draws from the notion that to effectively connect with diverse audiences and promote sales, it is essential for businesses to understand the cultural and ethnic contexts surrounding colors, languages, and iconography (Murray & Ozanne, 1991). The information extracted through the analysis methods illustrated in this thesis could be used in a number of ways. For urban planners, it could provide valuable insights into the economic health of different areas of the city and help inform decisions about future development. For sociologists, it could provide a rich dataset for studying the social and economic factors that drive changes in the commercial landscape over time.

Soja's analysis is highly theoretical and draws on a range of complex concepts and frameworks, which makes it less accessible to non-academic practitioners and policymakers interested in applying the book's insights to real-world problems. This thesis aims to take a more pragmatic and technical approach to postmetropolitan and regional analyses.

Additionally, the thesis proposes a framework for incorporating this data into urban planning and sociological research, providing a basis for further study in this area in the future, and potentially finding answers to questions such as:

- What industries are prone to forming clusters and why?
- What are the predictors of neighborhoods that are about to experience blight and high commercial turnover/exodus?

Precedents of Computer Vision as a Tool for Urban Analytics

In recent years, computer vision has emerged as an important tool for understanding urban change. As cities have witnessed a resurgence of social movements and public space protests, literature in sociology and communication has extensively examined the use of online public spaces, such as social media engagement, to support civic action and social movements. However, there exists a gap in understanding the relationship between online and offline public spaces and how they are culturally shaped by architecture and urban design to support these movements.

To address this gap, Cook's paper titled *Protest & Public Space: Challenging Notions of Sociospatial Distance in New York City* aimed to explore the nature of social movements in public spaces today, specifically focusing on Manhattan's Foley Square, which was utilized by three distinct social movements in January 2020. The study's findings suggest that a movement's online social network has palpable relationships with its tangible civic action in the form of public space protest. There are observable patterns across movements in the construction of themes of space, narrative, participating actors, and place. The insight gleaned from such patterns can greatly be enhanced with the added dimension of visual data.

Computer vision algorithms can provide valuable insights into the dynamics of urban environments, including patterns of economic activity, social inequality, and demographic changes, by analyzing high-resolution imagery.

Through the use of computer vision algorithms, large volumes of data can be analyzed in order to identify patterns and trends that might otherwise be obscure to human observers. By analyzing patterns of economic activity and land use across different neighborhoods and

regions, computer vision algorithms can assist researchers in identifying areas that are marginalized economically and socially.

For example, in Naik et al's 2017 paper, *Computer Vision Uncovers Predictors of Physical Urban Change*, a computer vision method is developed to measure and analyze changes in the physical appearances of neighborhoods from street-level imagery. The goal of this method is to identify the factors that predict neighborhood improvement and to better understand the dynamics of urban change. The results of this analysis show that neighborhoods with higher levels of education and higher population densities are more likely to experience improvements in their infrastructure, supporting existing theories of human capital agglomeration. Neighborhoods with better initial appearances are more likely to experience more substantial upgrading, as predicted by the tipping theory of urban change, and neighborhoods closer to city centers and other physically attractive neighborhoods experience more improvement, in line with the invasion theory of urban sociology.

Similarly in another study, titled *Visual Urban Sensing: Understanding Cities Through Computer Vision*, Naik introduces algorithms that are designed to compute quantitative measures of urban appearance and urban change. The first algorithm, called Streetscore, uses computer vision and crowdsourcing to quantify how safe a street block looks to a human observer. The second algorithm, called Streetchange, uses time-series street-level imagery to compute a metric for change in the built environment. A positive Streetchange is indicative of urban growth, while a negative Streetchange is indicative of decay.

The application of extensive data, including both tangible and intangible, has gained increasing popularity in urban planning due to its potential in providing valuable insights and improving decision-making processes. The results of these studies provide new insights into

the relationship between urban appearance, urban change, and various social and economic factors. In the context of this thesis, in which the focus is on storefront signages, the aim is to pinpoint commercial storefront signs from street view imagery– which computer vision techniques can help locate and identify.

In a 2016 paper, *Scene text detection and recognition: Recent advances and future trends*, Yingying Zhu et al discuss the challenges in detecting and recognizing scene text, including variability in rasterized text data representation, occlusion, and chaotic backgrounds. The authors point out that traditional feature-based methods such as histogram of oriented gradients (HOG) and scale-invariant feature transform (SIFT), have limitations in handling the variability of scene text. The paper endorses methods based on deep learning, particularly convolutional neural networks (CNNs), which have achieved excellence in many text recognition and recognition tasks while noting that character-based recognition methods are also quite effective but computationally intensive when used to recognize long text strings.

CNNs can be trained on a dataset of storefront signage images using labeled data to identify similar signs in the street view imagery, after which the model can be employed to detect similar signages. In contrast, word-based methods can efficiently recognize whole words, but may not be able to handle the variability of individual characters.

Optical Character Recognition (OCR) is another text-recognition approach that involves pre-processing the image to improve the quality of the text, segmenting the individual characters, extracting the features of the characters, classifying the characters as letters or numbers, and finally, recognizing the text– based on which the image may be classified.

Figure 1

Text Detection and Word Recognition



Note. The figure shows the word recognition results applied to a subset of the Street View Text (SVT) dataset (Visual Business Recognition – A Multimodal Approach – Center for Research in Computer Vision, n.d.)

One such method is the TextBoxes algorithm, which uses a single deep neural network to simultaneously localize and recognize text and achieves state-of-the-art performance on various benchmark datasets. In their 2017 paper, *TextBoxes: A Fast Text Detector with a Single Deep Neural Network*, Minghui Liao et al explain that the algorithm works by dividing an input image into a set of rectangular proposal regions, each of which is processed by a convolutional

neural network (CNN) to generate a set of feature maps. These feature maps are then used to predict the coordinates and confidence scores of text regions within each proposal region. The algorithm also includes a recognition module that uses a separate CNN to recognize the text within each detected region. One of the key advantages of the TextBoxes algorithm is its efficiency. Unlike many previous text detection algorithms, which require multiple stages and post-processing steps, TextBoxes can perform text detection and recognition in a single pass using a single neural network. This allows it to achieve high accuracy while also being computationally efficient.

The authors evaluated the TextBoxes algorithm on several benchmark datasets, including ICDAR 2013, ICDAR 2015, and MSRA-TD500. They found that TextBoxes outperformed several other state-of-the-art text detection algorithms in terms of both accuracy and efficiency.

In addition to the aforementioned frameworks, EAST and Tesseract are also popular algorithms used in the field of computer vision. EAST, or Enhanced Aspect-based Text detection, is a deep learning-based method for detecting text in images and video. It is able to detect text in arbitrary orientations and is able to handle a variety of fonts, sizes, and colors. Tesseract, on the other hand, is an optical character recognition (OCR) algorithm that is used to recognize and extract text from images. It is one of the most accurate OCR algorithms available and is widely used in a variety of applications.

Both EAST and Tesseract are widely used in applications such as automatic document analysis, image indexing, and automatic translation. This thesis will employ these algorithms for the purpose of Image Processing and Object Detection to extract Storefront Data and Positional Data (as referred to in Fig. 1).

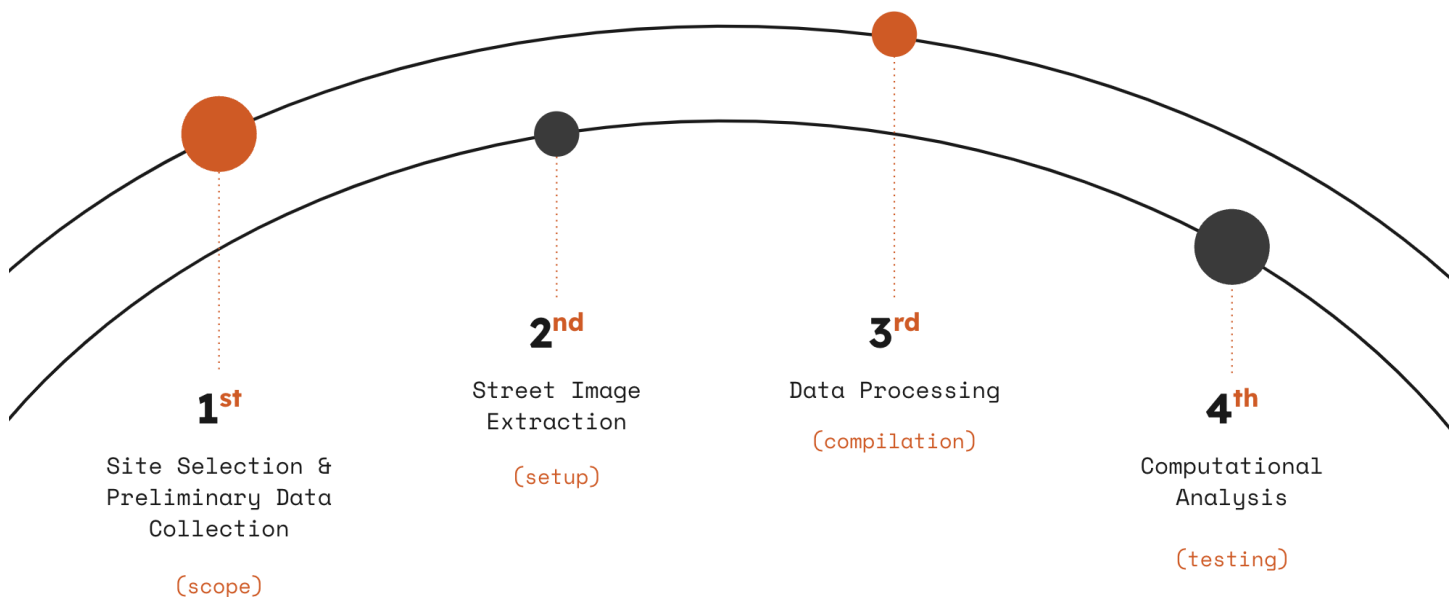
Research Question

In the context of this thesis, computer vision is used to analyze large amounts of visual data from cities, specifically street-level photographs, in order to identify changes in the commercial landscape over time. This includes the emergence of new businesses, the closure of existing ones, and changes in the types of commercial activities taking place in different areas of the city. This can potentially be achieved by extracting street view imagery to detect and extract visual information, namely colors and languages used across storefronts, to observe if and how they, and the changes they experience over time, reflect and contribute to the formation of enclaves in cities. Ultimately, this thesis aims to answer the question:

Can Computer Vision be used to gain insight into ethnoburbs & the changes they undergo?

Methods

The technical pipeline of the research can be broken down into four key steps.

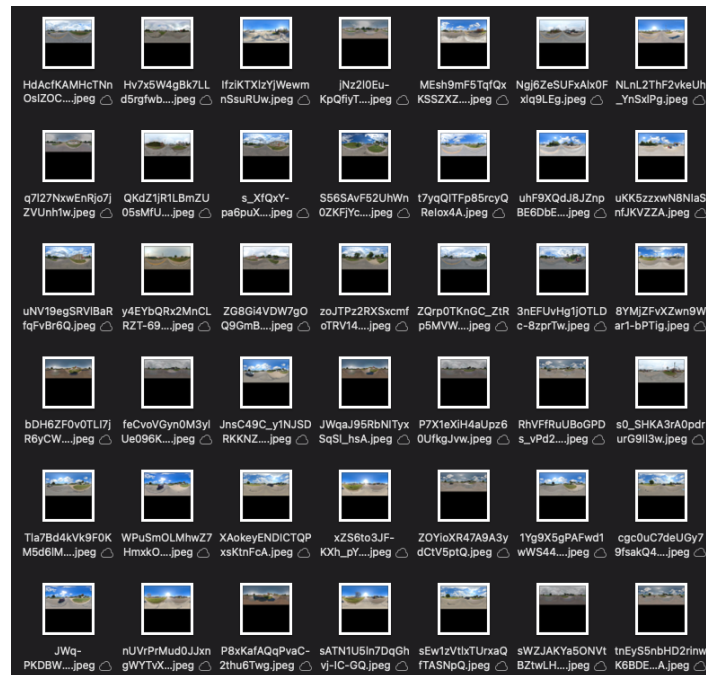


Data and Extraction

Street Image Data. Google Street View has a wide coverage of over 10 million miles of roads in more than 200 countries. Its user-friendly and well-documented API provides access to both still images and 360-degree panoramic views of locations, along with relevant metadata such as image capture time and location data. While Mapillary and OpenStreetCam also offer street-level imagery, albeit crowdsourced, they have more limited coverage and the quality of their images may be inconsistent depending on location and the contributor.

Figure 2

Extracted Street View Imagery (Subset)



Therefore, this thesis opts to use Google Street View as the primary source of street-level imagery.

Data Axle. Data Axle is a data solutions provider that offers comprehensive and accurate data on businesses, consumers, etc. This thesis uses Data Axle's Historic Businesses dataset, accessed via Wharton Research Data Services, which is a comprehensive collection of data that tracks the historical changes and trends in the business landscape in the United States.

Figure 3

Historic Business Data Subset

abi	archive_version_year	company	zipcode	primary_naics_code	naics8_descriptions	year_established	latitude	lon
6691505	1997	MONNIER INC	48001	44131022.0	AUTOMOTIVE PARTS & ACCESSORIES STORES	NaN	42.62827	-82.
8361735	1997	ALGONAC CAST PRODUCTS INC	48001	33151304.0	STEEL FOUNDRIES (EXCEPT INVESTMENT)	NaN	42.62988	-82.
142201151	1997	DOWNRIVER MEDICAL CTR	48001	62111107.0	OFFICES OF PHYSICIANS (EXC MENTAL HEALTH SPECS)	NaN	42.61620	-82.
142201185	1997	COLONY MEDICAL CLINIC	48001	62149301.0	FREESTANDING AMBULATORY SURGICAL & EMERGENCY CTRS	NaN	42.63572	-82.
142201201	1997	MC KOAN MC KOAN & MC KOAN	48001	54111002.0	OFFICES OF LAWYERS	NaN	42.61555	-82.

This dataset includes information on businesses that have operated in the US since 2003, including their names, addresses, phone numbers, industry classifications, employee size, revenue estimates, and other relevant data points. The query for the city of Detroit returned 5,642,124 rows, comprising 63,193 unique businesses.

This data is used to validate and train the computer vision models to detect businesses in images by providing a quicker way to label and annotate images, providing the model with examples of different types of businesses and their visual appearance.

Events. The anomalies and patterns that emerge in the analyses of the aforementioned data can be understood in further depth when quantitative data is used in conjunction with qualitative data, such as news media, which can provide valuable insight. This could include reports on changes in government policy, natural disasters, or other events that could have affected certain communities or industries— which could be narrated through interviews with individuals or businesses that have been affected by the decline, providing a more personal and human perspective on the data. Literature can also provide insight into the impacts of global phenomena such as the pandemic, which has had a widespread impact on businesses across the country, with many businesses forced to close or significantly reduce their operations due to government-imposed lockdowns and restrictions — leading to widespread economic disruption, with many businesses struggling to survive or adapt to the new circumstances.

The rise of e-commerce and imports has also had an impact on businesses as it has become easier for consumers to shop online and purchase goods from a wider range of sources, including international sellers. This has increased competition for businesses, particularly small and local businesses, which may struggle to compete with the convenience and lower prices offered by online retailers.

Changes in immigration policies have also had an impact on businesses, particularly those that rely on immigrant labor— namely industries such as agriculture, construction, and hospitality, which rely on immigrant workers to fill job positions. Changes in immigration policies, such as stricter enforcement or changes to visa programs, can make it more difficult for these businesses to access the labor they need, potentially impacting their operations and profitability.

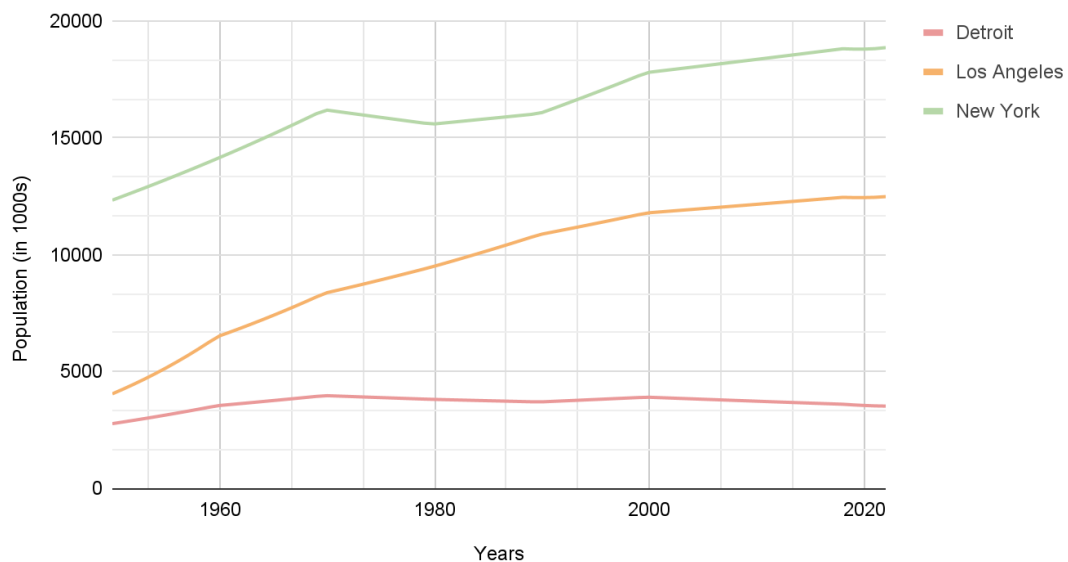
The thesis doesn't directly employ this contextual information as a part of the computational analyses but rather uses them to supplement the process of hypothesizing the causal and correlational relationships between socioeconomic phenomena and their effects on commerce and demographic change in ethnoburbs.

Site Selection Criteria

Detroit, New York, and Los Angeles are potentially good sites for this thesis. They are diverse and have rich cultural histories, with a variety of communities, traditions, and practices— and among the selected cities there is enough variation in population growth, economic activity, ethnic composition, and context to provide a rich dataset for ethnographic research.

Figure 4

Change in Population of Detroit, Los Angeles, and New York (1970–2022)

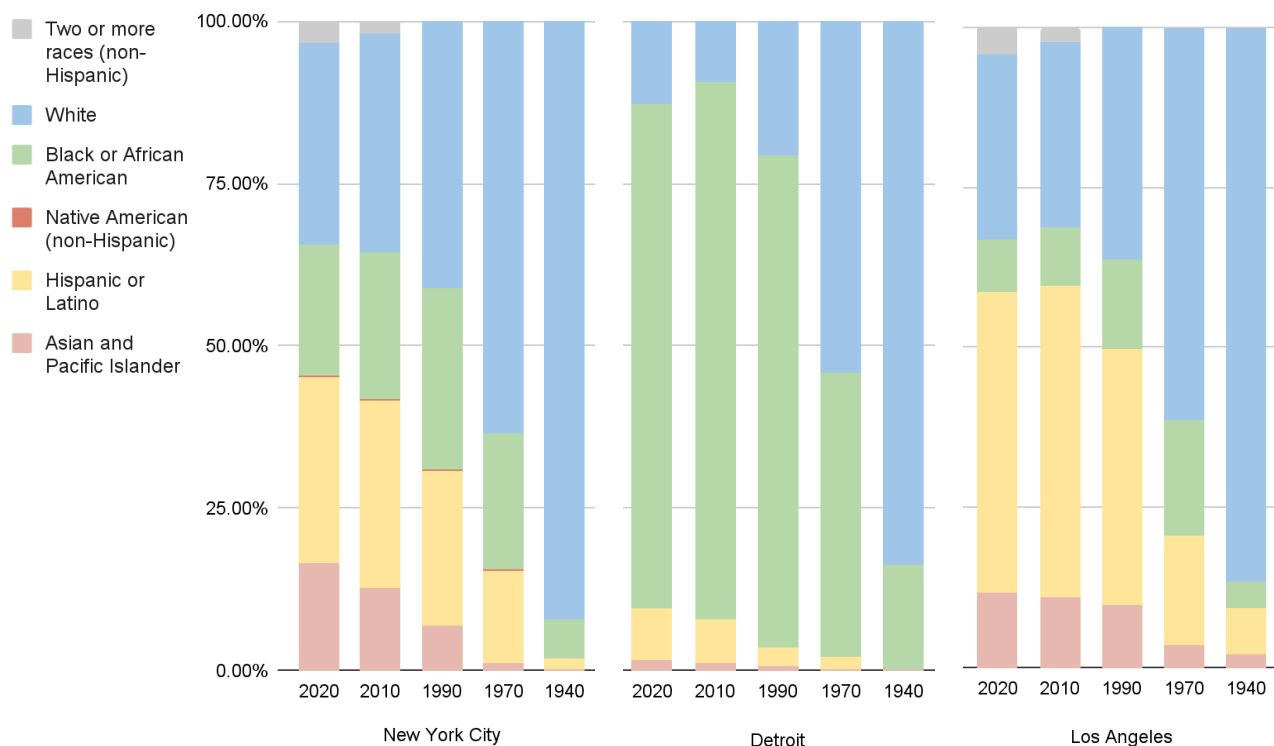


Note. Data collected from *World Population Prospects - United Nations*.

Detroit has a more industrial economy and is known for its automotive industry. It has a more homogenous population and is known for its working-class, blue-collar culture. New York City, on the other hand, is known for its diverse population and its vibrant arts and cultural scene. It is home to many different ethnic and racial groups, and it is a major global hub for finance, business, and media. Los Angeles has been known to historically attract immigrants from around the world, especially Latin America and Asia leading to the emergence of distinct ethnic enclaves.

Figure 5

Change in Racial Composition of Detroit, Los Angeles, and New York (1940–2022)



The sites also have very different urban landscapes. New York City is known for its dense, high-rise buildings, bustling streets, and iconic skyline. Although similar to New York in cultural capital, Los Angeles' spread out and decentralized urban form— among other facets— has resulted in a drastically different subculture. Detroit, on the other hand, is known for its sprawling, low-rise buildings and its wide, open streets. It has a more industrial feel.

Understanding the context of these cities is an essential part of this thesis as it forms the basis for all hypotheses that arise from the computational analysis and the patterns that emerge from it.

Detroit. Detroit has experienced significant changes in its commercial landscape and ethnic composition over time, making it an ideal site for this thesis. As a historically industrial city that has undergone significant changes in its economic structure over time, the city's economic decline since the 1950s has had a profound impact on its commercial landscape— resulting in the closure of many businesses and the emergence of new ones.

The city's history of migration began in the early twentieth century when African Americans from the South migrated to Detroit— as a part of the Great Migration— in search of employment opportunities in the booming automotive industry (Sugrue, 2014).

Additionally, it has a diverse ethnic population, including African Americans, Arab Americans, and Asian Americans. Many Arab Americans arrived in Detroit from Syria, Lebanon, and Palestine during the early twentieth century (Lee, 2005). These immigrants were largely Christian and were attracted to the city by employment opportunities in the auto industry, as well as the presence of other Arab Americans in the area. Over time, the Arab

American population in Detroit grew, and today, Detroit is home to one of the largest Arab American populations in the United States (Li, 2001).

The city's ethnic composition has changed significantly over time due to immigration and migration (Sugrue, 2014). In particular, the history of Asian migration to Detroit dates back to the 19th century, with the arrival of Chinese immigrants who settled in what became the Chinatown neighborhood, which was followed by the influx of other Asian groups, including Japanese, Korean, and Vietnamese migrants who migrated to the city in the latter half of the twentieth century. The first wave of Japanese immigrants to Detroit arrived in the early twentieth century, with many working in the city's automotive and steel industries. However, the majority of Japanese migration to Detroit occurred after World War II, when Japanese automakers began to establish operations in the city. Korean and Vietnamese immigrants arrived in Detroit in the latter half of the twentieth century, with many opening small businesses in the city's urban neighborhoods (Lee, 2005).

Economic conditions, such as the decline of the automotive industry in Detroit, have played a significant role in the closure of many businesses and the emergence of new ones. Immigration policies, such as the Immigration and Nationality Act of 1965, have also influenced the city's ethnic composition by opening up opportunities for immigration from non-European countries.

Los Angeles. The city of Los Angeles has experienced drastic shifts in its ethnic makeup over the course of several years. The contributing factors to this diversity include various influxes of migration spurring changes in demographics. For instance, the city's Latino population grew from 10% in 1940 to 48% in 2019 (U.S. Census Bureau, 2020). Along

those same lines lies an increase among Asian community members; from numbering less than one percent eight decades ago to fully occupying twelve percentage points more recently (U.S Census Bureau, 2020). These changes in ethnic composition have led to the emergence of ethnic enclaves, where businesses and cultural activities reflect the heritage of specific communities.

The commercial landscape in LA has also undergone significant changes over time. During the twentieth century, commerce was chiefly conducted in downtown Los Angeles; its grandeur was marked by massive department stores, banks, and theaters. However, during the 1950s-1960s suburban expansion took place rapidly on account of developing freeways which made businesses spread out (Scott, 1997). Simultaneously with this particular dispersal came growth among ethnic communities along specific parts known for their small specialized businesses to appeal to these segments' demands.

An array of factors has contributed to LA's evolving ethnic compilation and commercial panorama. Immigration policies, economic prospects, as well as social networks have all had their part in shaping the city's formation. The Immigration and Nationality Act of 1965 facilitated the entry of immigrants from Asia and Latin America, leading to the growth of these communities in LA (Wu, 2015). Commercial changes were also brought about mainly due to a decrease in manufacturing activity combined with an increase in noteworthy growth in service sectors that affected distinct regions throughout the city over time (Scott,1997).

The relevance of Asian migration in LA has had a significant influence on creating ethnic enclaves and diversifying the commercial sector. Initially, Chinese migrants came to work in mines and agriculture during the 19th & early 20th centuries (Wu,2015). The restriction imposed on Chinese immigration due to The Chinese Exclusion Act of 1882

resulted in Japanese, Filipino, and Korean communities increasing in number within Los Angeles (Wu,2015); each group established its own businesses and neighborhoods which eventually led to Little Tokyo, Historic Filipinotown, Koreatown among many other places.

These neighborhoods are not only significant for their cultural and historical importance but also for the diverse communities that call them home.

New York. Over time, the economic activity and population growth in New York City have pushed significant changes to its commercial landscape. Industrial areas have transformed into lively districts featuring small businesses, restaurants, and cultural institutions throughout many neighborhoods. Neighborhoods such as SoHo and Williamsburg have undergone gentrification and now attract an array of boutique shops and art galleries, while neighborhoods such as Harlem have seen a resurgence of investment and business activity after decades of disinvestment.

Home to over 8 million people, New York City's population has been shaped by waves of immigration from diverse cultural backgrounds. As of 2020, the city's population is composed of Whites (42.7%), Hispanics or Latinos (24.3%), Black or African Americans (24.1%), and Asians (13.9%) (U.S. Census Bureau, 2020). These diverse communities have contributed to the city's thriving economy and rich cultural landscape, which is reflected in its commercial districts.

Asian migration, in particular, has played a significant role in shaping the city's commercial landscape. The first wave of Asian migration to New York City occurred in the mid-19th century with the arrival of Chinese immigrants who worked on the transcontinental railroad and settled in areas such as Chinatown (Waxman, 2007).

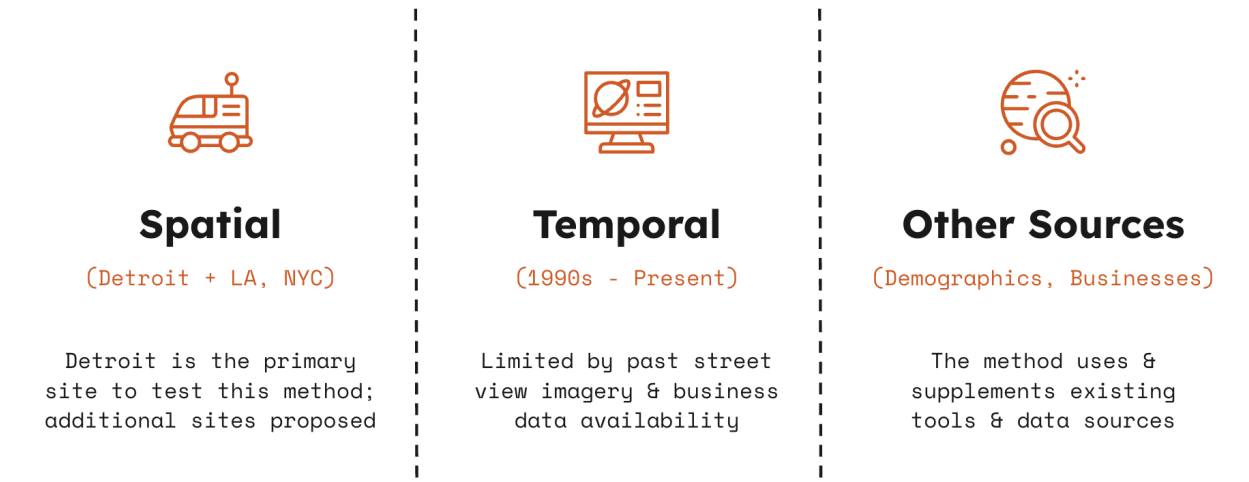
Today, neighborhoods such as Flushing and Sunset Park are hubs for Asian-owned businesses and cultural institutions. The Asian population in New York City has grown significantly in recent years, from 1.1 million in 2000 to 1.5 million in 2019.

"The Korean enclave in the Flushing area is a truly multiracial enclave. Whites, Chinese, and Hispanics, respectively outnumber Korean Americans by a large margin. This multiracial characteristic of the neighborhoods gave Korean immigrants a disadvantage in electoral politics and other empowerment areas." (Min, 2023)

In *The Advantages of Suburban enclaves over urban enclaves for community empowerment: Korean immigrants in greater New York*, Min compared the ways in which a single ethnic group, Korean Americans, have fared in differing cities in the same region. While Korean enclaves exist in both urban and suburban New York, the percentage of the population they represent in these two spaces differs. As such, in the suburb of Bergen County, where Korean Americans are a large chunk of the community, they tend to do better in electoral and cultural politics than do Koreans Americans in Queens. A suburban enclave does not guarantee sociopolitical domination, however, as in Flushing, where Chinese Americans outnumber Korean Americans, the Korean American community holds less clout.

The economic and commercial landscape changes in New York City have been influenced by various factors such as globalization, which has resulted in increased economic competition and has led to the growth of certain sectors such as finance, media, and technology. Government policies and initiatives have also played a role, such as zoning regulations, tax incentives, and public investment in infrastructure (Storper, 2013).

Defined Scope



Data Limitations

Although the Google Street View API is a valuable resource to this thesis, it does have one significant limitation— the API only provides panoramas for the latest image capture for a particular set of coordinates, making it difficult to compare it with past street view imagery. This is overcome by deploying web-scraping scripts to identify and extract the imagery for the years that the API doesn't provide support for.

The Historic Businesses Dataset, although quite comprehensive, does not include information on very small or new businesses that have not yet been established for long. Additionally, the data may not always be up-to-date, as businesses may have changed their information or gone out of business since the data was last collected. This thesis hence uses it in conjunction with manual verification and labeling to verify the accuracy of the data before feeding it into the training algorithms.

Computational Analyses

Data Collection. To collect a large dataset of street-level photographs from different areas of the city, taken at different points in time, the first step is to verify if there is a Street-View image available for a certain set of coordinates. In the case of Detroit, the availability of imagery varies depending on the year and location. Google's metadata API is used to extract this information through a simple function:

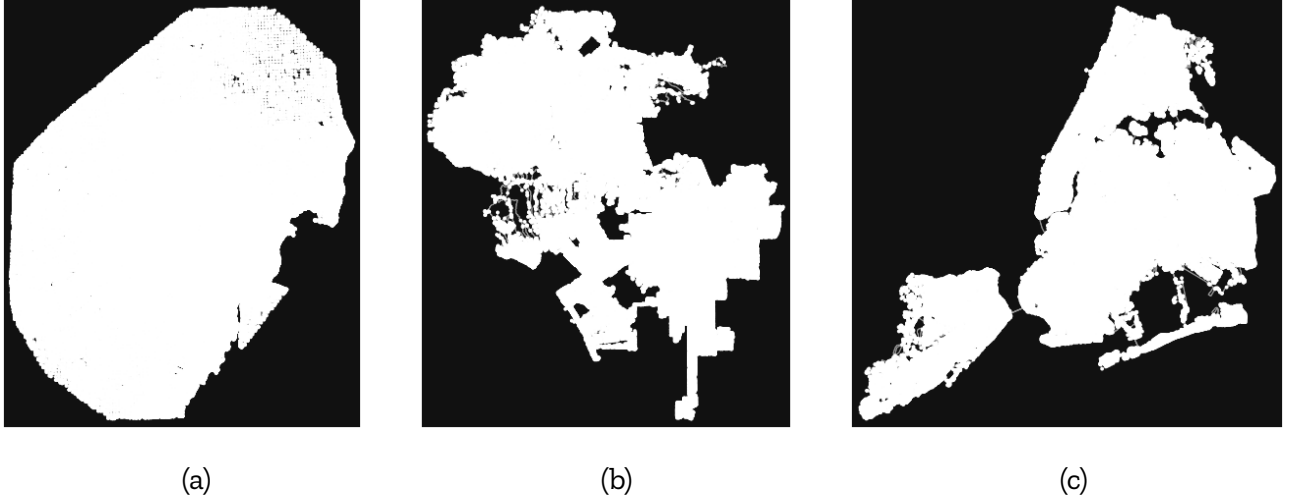
```
def get_metadata(lat, lon):
    url =
    "https://maps.googleapis.com/maps/api/streetview/metadata?location={0:},{1:}&key=
    AIzaSyC7AyrDN9jjs6WP4l_6frUGcjFpXqc4Edg"
    response = json.loads(urlopen(url.format(lat, lon)).read())
    if response['status'] == 'OK':
        url =
        'https://www.google.com/maps/@{0:},{1:}/data=!3m7!1e1!3m5!1s{2:}!2e0!5s20201001T0000
        00!7i16384!8i8192'
        stview = url.format(lat, lon, response['pano_id'])
        return response['status'], response, stview
    else:
        return response['status']
```

The `osmx` (Boeing, 2017) package is a Python library that retrieves OpenStreetMap (OSM) data, including GPS coordinates of buildings, streets, and points of interest. In this thesis, it is used to collect all the street nodes of a city— after which the aforementioned function is used to validate the presence of panoramas and collect information about the year and month of upload.

Nearby panoramas are collected as well using a function that interpolates the points every 10-20 meters between every two nodes. This provides the most comprehensive coverage of the potential locations that have a street-view photo available.

Figure 6

Node Maps generated using osmx for (a) Detroit, (b) Los Angeles, and (c) New York



The points' metadata is collected using the Google Street View metadata API, and is then stored in a database. However, this API generally only returns the nearest point's metadata. To ensure completeness, the locations and unique IDs for the adjacent street views that the original node's link points to is also run through the API call. Throughout this process, about 10.3% of all calls have had **'NO RESPONSE'**. These locations are often private property or inaccessible areas.

The *panoid* field refers to the unique ID assigned to each Street View Panorama available on Google Maps. This thesis exclusively uses images extracted from *panoids* that have a valid non-zero corresponding year and month. This helps avoid user-generated images that may have validation issues and duplicate imagery.

Figure 7

Comparison of Google Street View Imagery Availability in Detroit (2007-2020)

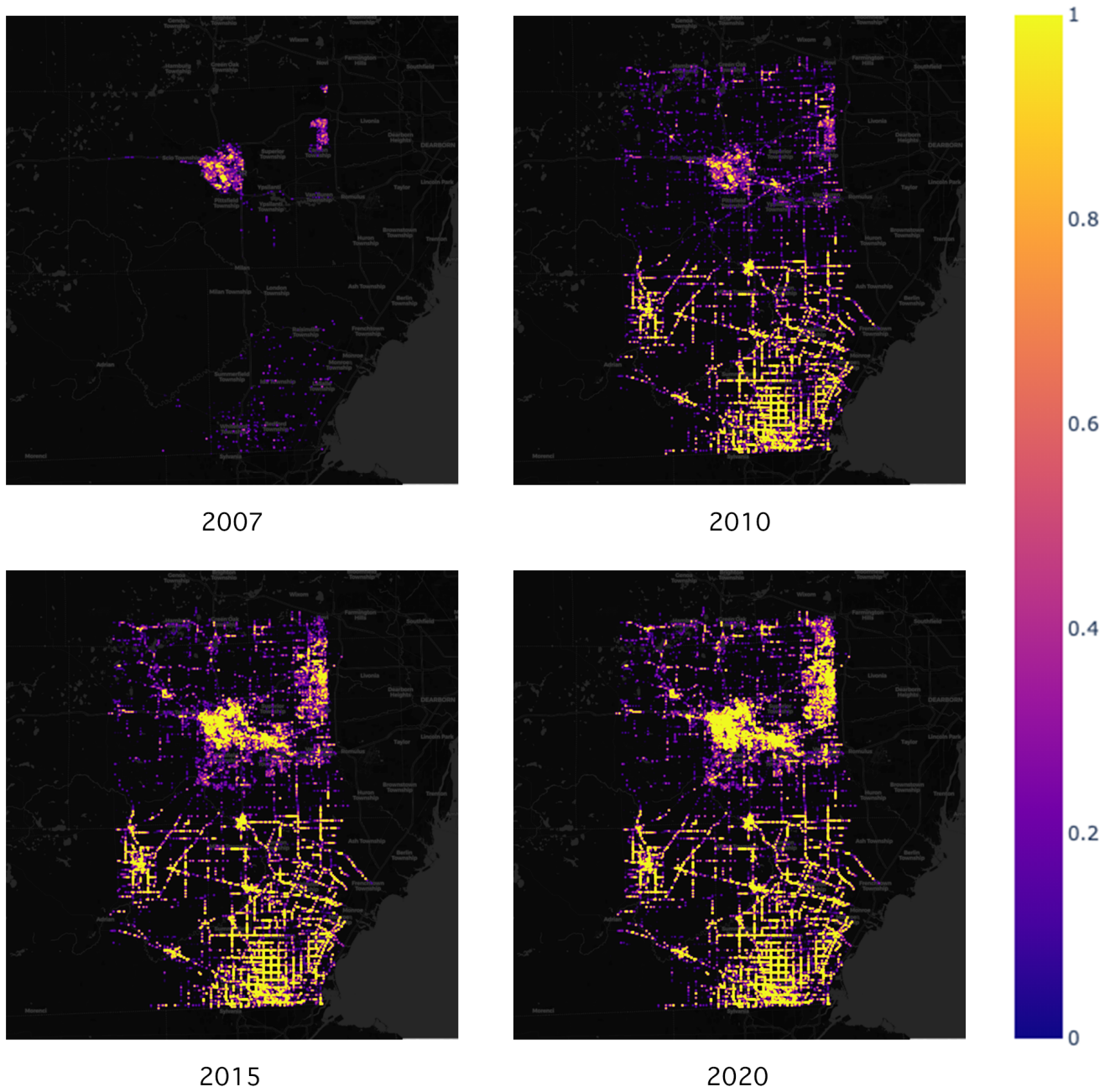


Table 1

The resulting table for a sample query where the input latitude and longitude are 42.380506 and -83.796114

	panoid	lat	lon	year	month
0	lc14HV-rts7e8pXM7k8v2Q	42.380494	-83.796151	2008	10
1	S-pBG1lNpf9VWlnNikM6kw	42.380483	-83.796161	2011	9
2	zIFExQGMz-ZveJftFlJckw	42.380489	-83.796183	2015	9
3	63JrH4FUI386A8maIYm6fw	42.380486	-83.796142	2016	9
4	q9rsj4XJ-L80rH4q1IBlaA	42.380495	-83.796624	2020	11
5	uOWK2wbMJHrUJSVrN-TYBg	42.380486	-83.796151	2021	6
6	j_ISNO-JZmFRMhkjK3DNzw	42.380485	-83.796002	0	0
7	j_CNo9DYMJqtTXpOmLaAbw	42.380526	-83.796323	0	0
8	k-nevVrFmF9Zwa0cqBKfgA	42.380511	-83.795563	0	0
9	TexmaomTqrvEmGltTRu4wg	42.380479	-83.795685	0	0
10	SIyp680nI3bdo71gcPKfMg	42.380482	-83.795844	0	0
11	5wMCVu0I_EzrEmcc0eNlXg	42.380528	-83.796491	0	0
12	IWIRQicv-ySsgoTEONWcdQ	42.38054	-83.796184	0	0

All invalid rows are then run through the pre-extrapolation function as a secondary verification method to ensure that no field has been marked invalid in error. The final consolidated database is used to extract the relevant imagery.

The next step employs Google Maps Javascript API's StreetViewPanorama, a component that delivers comprehensive 360-degree views of a particular site or location, which utilizes images that adhere to the equirectangular (Plate Carrée) projection (*Street View Service / Maps JavaScript API*, n.d.). This projection captures a complete wrap-around of 360 degrees in the horizontal view and 180 degrees in the vertical view. Consequently, the resulting image has an aspect ratio of 2:1, depicting a seamless panoramic view.

Figure 7

Panorama and Tile Composition



Note. (Street View Service | Maps JavaScript API, n.d.)

This process is intensive, both in terms of time and computation power due to the sheer volume of data, and the limits placed on API calls and direct downloads.

Detroit alone generates 3,561,914 street-view images.

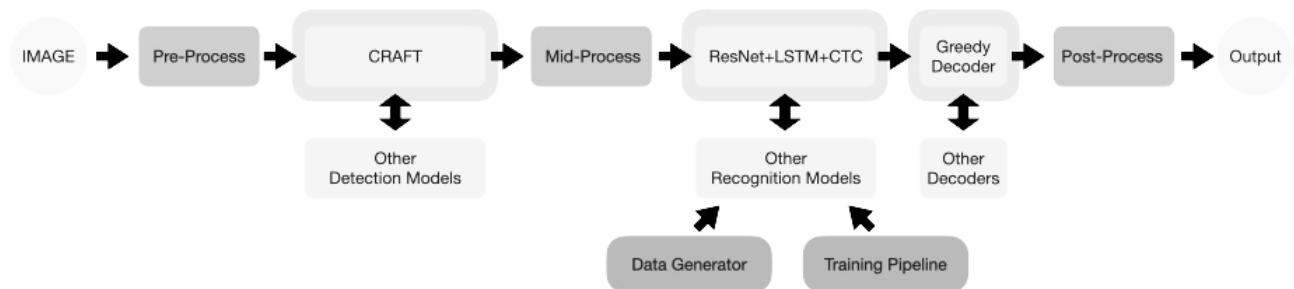
Text Extraction. Once the street-level images have been extracted, it is necessary to detect and recognize the text contained within the images. Although scene text detection frameworks like OpenCV can easily identify Latin-script languages, such as English, Spanish, and German, the ability to determine the language of the text, regardless of script, is crucial for this thesis.

To address this issue, this study utilizes EasyOCR (*Jaied AI: EasyOCR Documentation, n.d.*), which is a Python OCR module capable of reading natural scene text as well as dense text in documents. With support for over 80 languages, EasyOCR is an excellent choice for this particular use case.

The algorithm doesn't operate losslessly, and because of the size of the stitched panoramas, directly inputting the images can result in highly pixelated photos that become unreadable by the OCR. For this reason, it is essential to pre-process the images– by using OpenCV to convert the images to greyscale and improve the contrast– and then crop them to the text boundaries.

Figure 8

Easy OCR Framework



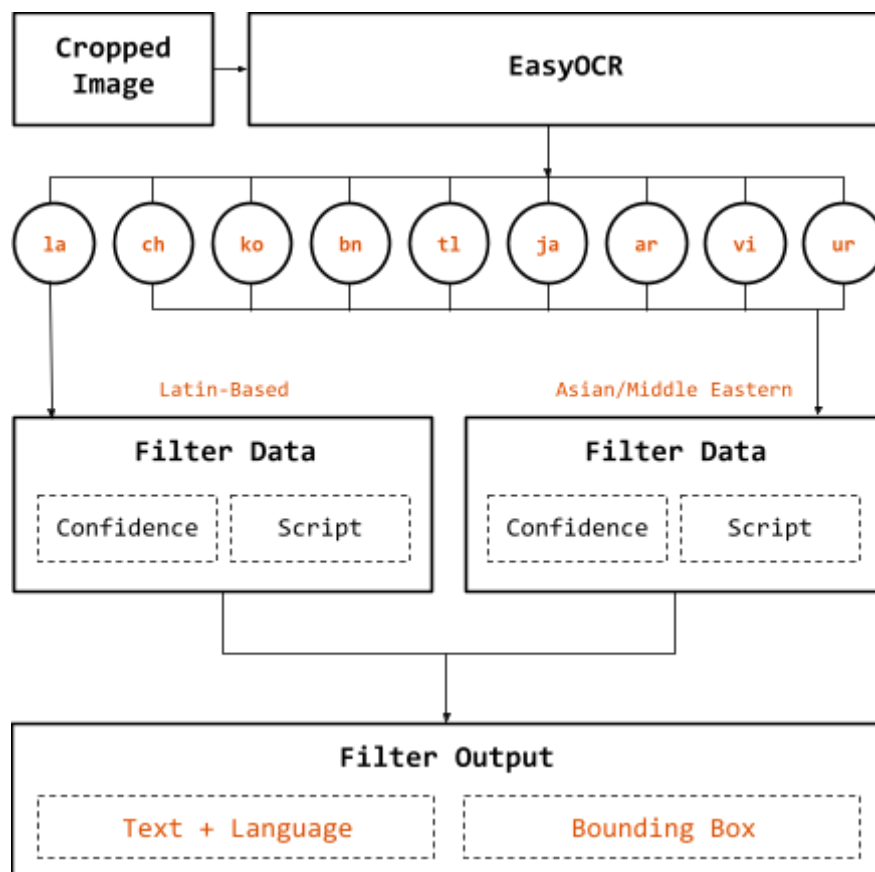
Note. Figure from EasyOCR's Github (EasyOCR, 2020/2023)

In natural scene images, this becomes more challenging as stroke width algorithms struggle to tell the difference between scene text and other high-contrast elements. Stroke width algorithms are used to analyze digital images containing text, and OpenCV is a widely used framework for implementing them. These algorithms rely on the consistent stroke width of text characters to distinguish them from other parts of an image. There are various methods for implementing stroke width algorithms, such as computing the distance transform or gradient image of the original image. OpenCV provides functions such as morphology operations and gradient computation to implement these algorithms. However, in

Google Street View Images, the panoramic images generally contain the subject matter in the middle third of the image, horizontally. Cropping the image to this boundary is sufficient for this pipeline, although the stroke width algorithms will come into play later on.

Figure 9

Text Detection and Recognition Pipeline



Note. The shorthands for languages used above refer to Latin scripts (la), traditional Chinese (ch), Korean (ko), Bangla (bn), Tagalog (tl), Japanese (ja), Arabic (ar), Vietnamese (vi), Urdu (ur).

Language Analysis. In order to identify the language of the text that is detected in street-level images, a language detection model is employed for each language. The cropped image is then passed through each language model to determine the language of the text. The confidence level threshold is set to 60% or higher in order to achieve the best results. The output is then analyzed to determine the presence of ASCII script, which indicates that the text is in a Latin script-based language such as English, Spanish, French, or German. Conversely, the Asian/Middle Eastern language models exclude any results that pass the aforementioned check.

The text language is cross-verified using *langdetect*— a language detection library that is commonly used in natural language processing applications. It is an open-source Java-based library that can detect the language of a given text based on a statistical analysis of character n-grams, which are contiguous sequences of n characters. The algorithm of Langdetect is based on a trigram model that calculates the likelihood of a text belonging to a specific language by comparing the frequency of trigrams in that text to a frequency distribution of trigrams for each language. This method has been shown to be highly accurate in detecting the language of a given text in a variety of contexts.

Langdetect supports a wide range of languages, including most European and Asian languages, and can detect up to 55 different languages. The library provides an easy-to-use API that allows developers to quickly integrate language detection functionality into their applications. Additionally, Langdetect can handle multilingual text, where the text contains multiple languages, by returning the language with the highest probability.

The final combined outputs of the pipelines of the two script categories present themselves as an array similar to as follows:

```
[
  [[2415, 696], [2500, 696], [2500, 759], [2415, 759]], '바', 0.6713358944890366, 'ko'],
  [[5216, 3885], [5360, 3885], [5360, 3935], [5216, 3935]], '岬,卅刑', 0.886191203924262, 'ch']
  [[2415, 696], [2500, 696], [2500, 759], [2415, 759]], '川', 0.9892983995144142, 'ja']
  [[7275, 3562], [7908, 3562], [7908, 3748], [7275, 3748]], 'HMART', 0.6297556503409533, 'la']
]
```

The array consists of arrays formatted as *coordinates*, *text detected*, *confidence*, and *language*.

Figure 10

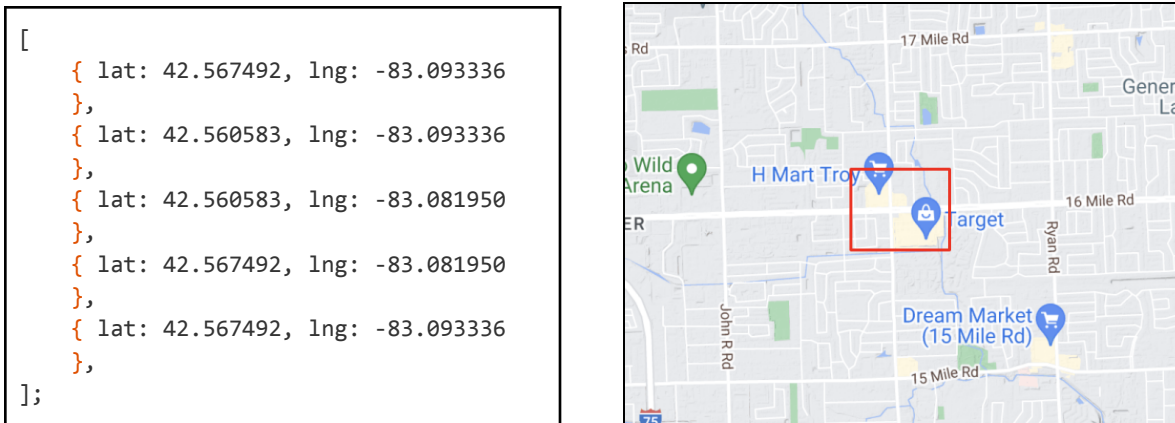
Change of Single-Race Categories in Macomb County, MI (2015 vs. 2020)



The bounding box arrays are stored and used to process the image further. For the purpose of testing the code and feasibility of the model, a sample boundary containing two local strip malls is taken from the larger Detroit region:

Figure 11

Change of Single-Race Categories in Macomb County, MI (2015 vs. 2020)



The bounds contained 9,237 valid panorama IDs, out of which 8,734 corresponded to valid images. After manual cleaning to remove images that contained vehicles with text decals that could potentially skew the data, images of residential or barren areas without commercial signages, and images with advertising that isn't attached to a proximate establishment, the resulting dataset comprised of 633 images. High resolution imagery was available for 577 of them. The top (sky) and bottom (ground) 25% of the image were cropped out, as the site primarily comprises of low-rise structures.

Findings. 3.7% of text with over 60% confidence was in Arabic, followed by Chinese at 2% and Korean at 1.3%.

Figure 12

Easy OCR text detection output for detecting English and Arabic Text

	Text	Confidence
0	ل!با ددا 31	0.0268
1	I FFJTIII	0.0521
2	54	0.7314
3	ACCOUMNG	0.3021
4	5٧٥٥	0.1114
5	LAIم	0.0712
6	MORIACI	0.1076
7	-SEPMICI5	0.084
8	[S141	0.0755
9	S٧VICLS	0.0976
10	خدمات عنارية	0.5478
11	خدمات	0.2706
12	عذاريا	0.2701
13	نس4	0.1243
14	٥	0.0489
15	أما م	0.0421
16	50097:0	0.3073
17	٤٤٧٩٧ ٢٥٧	0.0027



Figure 13

Easy OCR text detection output for detecting English and Korean Text

	Text	Confidence
0	어 제t	0.0465
1	M	0.1979
2	'IImMI	0.0439
3	"기	0.044
4	FIT0-	0.0223



Although EasyOCR has been shown to have high performance in multilingual scene text detection, it may still mislabel certain languages that use traditional Chinese characters, such as Japanese. This is because characters in these languages are more complex and exhibit greater variation in shape than those in Latin-based languages such as English or Spanish. This intricacy presents a challenge for OCR algorithms to accurately recognize characters and differentiate between them. As a result, the algorithms may mislabel or incorrectly identify the language.

Moreover, OCR algorithms may not be able to detect the entire text in an image, which could lead to the incomplete text being passed through a langdetect model for language verification. This, in turn, can result in false positives or inaccurate language identification.

Manual validation is often required to overcome these limitations. However, manual validation can be a time-consuming process, especially when dealing with large volumes of data. Crowd-sourcing translation or validation could be an option in the case of a public-facing output.

Color Analysis. After the text is extracted, the next step is to analyze the language and color of the storefront signs. OpenCV is used to extract cropped images from the bounding box coordinates. Unlike in the previous step, unprocessed images are used in this step to avoid data loss. *Color Thief* (Color Thief, n.d.) is a package in Python that makes it easy to extract colors from an image. It uses the K-means algorithm to group pixels in an image and chooses the most typical color of each group to determine the dominant colors of the image. After generating a palette of colors for each cropped image, the results are stored along with coordinates.

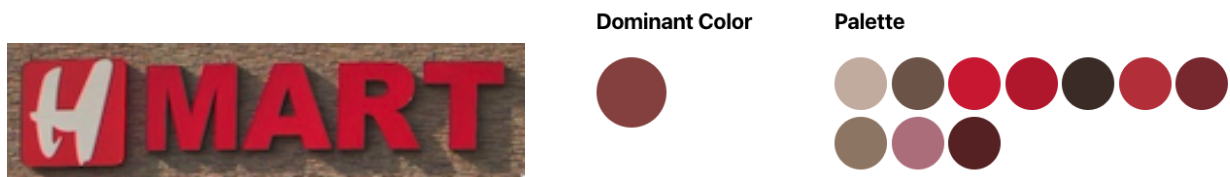
Figure 14

Bounding Box Extraction



Figure 15

Color Thief Analysis Sample Result



Findings. The results of this analysis do provide a comprehensive palette of colors. However, many of the colors tend to be background colors– including shadows and texture. This skews the color data and additional pre-processing might be required, perhaps by using OpenCV to grey out the background.

Data Analysis. Once the language and color information is extracted, the final step is to analyze the data.

Figure 16

Historic Business Data (1997-2020)



This involves clustering similar signs based on language and color, analyzing trends in the emergence and closure of businesses, and identifying changes in the types of commercial activities taking place in different areas of the city. The data extracted through this analysis can be validated using the Historic Businesses Dataset, which is also used to find the dominant industries in ethnic enclaves, as shown in fig. 16.

The research concludes with the following findings:



1.

Street view imagery **can** be used to extract visual data



2.

It needs to be used **in conjunction with** other sources



3.

Linguistic data can be a reflection of local demographic change



4.

Many more years of historic imagery is needed to track **substantial** change



5.

Self-orientalization & rebranded gentrification **aren't detected**

Results

Ideally, the output would be a digital tool that tells the story of ethnic enclaves in cities and maps the quantifiable changes in their characteristics over time, supplemented by

qualitative data to draw causal inferences/hypotheses between said changes and events (local, and global, social, virtual, etc.) that could have had an impact.

Public Facing Output. An informative interactive interface comprising of temporal maps of three cities, with different informational layers such as language, colors, industry, and demographic composition would help the public gain a better understanding of the way their cities have changed over time and the impact and footprint left by immigrants in the United States. The historic erasure and mislabeling of immigrant demographics make this a significant task in itself.

Planner/Technologist Facing Output. The methodology used in this thesis can provide a scalable framework for technologists and urban planners to understand the context of neighborhoods at a human-scale. This is becoming increasingly important in the age of Big Data Analytics, where a lot of the geospatial inferences gained are at a macro level that doesn't often give much insight into the nuances and local contexts within cities.

Potential Implications

The potential implications of developing this framework for urban planners to understand trends in neighborhood demographic change are numerous. For one, this could help urban planners to make more informed decisions about how to allocate resources and plan for the future growth and development of a given neighborhood. By providing urban planners with detailed and accurate information about current and projected demographic

trends, this tool could help them to identify potential issues and challenges that may arise, and take steps to address them in a proactive manner.

This thesis also acknowledges that there are potential risks associated with the use of visual big data. One of the most significant dangers is the potential for biases in the data, which could lead to negative impacts on marginalized communities. For example, algorithms based on visual data in predictive policing could result in discriminatory outcomes and disproportionately target people of color. Incomplete or inaccurate information is also a possibility. Relying exclusively on satellite imagery and physical design data could lead to incomplete and even misleading conclusions about a neighborhood's social and economic characteristics. Certain tools could also pose concerns about privacy and surveillance, which could infringe upon the privacy rights of individuals and communities. Moreover, the use of Big Data in participatory planning could lead to challenges in interpretation for individuals without technical expertise, thus resulting in a lack of meaningful engagement.

In research that involves computational sociology, such as this thesis, it is important to consider the limitations and potential risks to ensure that it is used responsibly and equitably.

Importance in Planning

This research holds particular significance for Asian immigrants in the US, who face unique challenges as they seek to establish themselves in a new country. Understanding the cultural nuances and preferences of Asian communities can help businesses establish cultural relevance and forge deeper connections with consumers. By using computer vision technology to extract information about the commercial landscape of a particular area, we can provide

valuable insights into the economic health of different areas of the city, which can be used to inform decisions about future development.

Moreover, the data extracted through this research can be used by sociologists to study the social and economic factors that drive changes in the commercial landscape over time. It can provide a rich dataset for studying the cultural and ethnic contexts surrounding colors, languages, and iconography that are critical for businesses to effectively connect with diverse audiences and promote sales.

// Analysis

Understanding
neighborhoods & unseen
cultural parameters

// Prediction

Ethnic enclaves,
gentrification, &
displacement

// Scalability

Reduced need for manual
site analysis & studies

// Archiving

A form of visual/oral
historic preservation

By providing urban planners with a better understanding of the needs and preferences of the people living in a given area, this tool could help them to design and implement policies and programs that are more responsive to the needs of the community. This, in turn, could help to improve the overall quality of life in the neighborhood, by making it a more welcoming, inclusive, and livable place for all residents.

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